125101 - Analytical chemistry 1 for engineers

Tutorial 8 – Precipitation/Redox Titration

T8-1. Consider the titration of 50 mL KBr (0.1 M) with $AgNO_3$ (0.2 M):

$$Ag^+ + Br^- \longrightarrow AgBr\ (s) \quad K_{sp}\ (AgBr) = 5.0\times10^{-13}$$

Find $p[Ag^+] = -log\ [Ag^+]$ for the following titrant volumes:

a. 15mL

b. 25mL

c. 30mL

T8-2. What is the concentration of CrO_4^{2-} that should be added to solution in Mohr titration of the followings halides with Ag^+?

K_{sp} (AgCl) = 1.8×10^{-10}

K_{sp} (AgBr) = 5.0×10^{-13}

K_{sp} (AgI) = 8.3×10^{-17}

K_{sp} (Ag_2CrO_4) = 1.2×10^{-12}

T8-3. 30 mL of a solution containing an unknown concentration of Cl^- was added to 50 mL of $AgNO_3$ with a concentration of 0.365 M. Later, the precipitate, $AgCl$, was filtered from the solution. The indicator, Fe^{3+}, was added to the filtered solution and the solution was titrated with KSCN with a concentration of 0.287 M. After the addition of 37.6 mL of the titrant, the solution turned to a pale rose color. What are the concentration of the Cl^- in the pristine solution?

The reactions:

$$Ag^+ + Cl^- \rightarrow AgCl(S) \quad K_{sp}(AgCl) = 1.8 \times 10^{-10}$$

The excess Ag^+ was titrated by KSCN:

$$Ag^+ + SCN^- \rightarrow AgSCN(S) \quad K_{sp}(AgSCN) = 1.2 \times 10^{-12}$$